A metacommunity by nested composition

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```
ENV["GKSwstype"] = "100"
using MatrixCommunityModels
using Catlab: @relation
using Plots, Statistics
```

Why nest

Lytle & Tonkin (2023) point out that an isolated matrix community model tends to lose species, and that placing the habitat in a metacommunity "deserves further attention". Because a composed community is itself something that can be composed, a metacommunity needs no new machinery: patches are communities, and dispersal is one more kind of pool.

Species that export seed

A grass and a shrub compete for space within a patch. Establishment of seedlings is limited by the free space in that patch. A fraction `m` of each species' seed leaves the patch into a seed rain shared by all patches; each of the two patches receives half of it. A species therefore has two ports: its patch's `space`, and its own seed rain.

Patch `B` burns periodically, which kills most shrubs and some grass. The environment of each year says which patches burn.

```
K = 1000.0

function grass(patch; m)
    crowding(ctx) = max(0.0, 1 - ctx.pool.space / K)
    burns(ctx) = getfield(ctx.env, patch)
    Species(Symbol(:grass_, patch), [:seedling, :adult],
            [survival(:establish, :seedling => :adult, ctx -> 0.30 *
 ↪  crowding(ctx)),
             survival(:persist, :adult => :adult, ctx -> burns(ctx) ? 0.35 :
              ↪  0.50),
             fecundity(:seed, :adult => :seedling, 6.0 * (1 - m))];
            occupancy = [:space => [0.0, 1.0], :grass_rain => [0.0, 6.0 *
 ↪  m]],
            recruitment = [:seedling => ctx -> ctx.pool.grass_rain / 2])
end

function shrub(patch; m)
    crowding(ctx) = max(0.0, 1 - ctx.pool.space / K)
    burns(ctx) = getfield(ctx.env, patch)
    Species(Symbol(:shrub_, patch), [:seedling, :juvenile, :adult],
            [survival(:establish, :seedling => :juvenile, ctx -> 0.30 *
 ↪  crowding(ctx)),
             survival(:mature, :juvenile => :adult, ctx -> burns(ctx) ? 0.05
              ↪  : 0.50),
             survival(:persist, :adult => :adult, ctx -> burns(ctx) ? 0.10 :
              ↪  0.95),
             fecundity(:seed, :adult => :seedling, 3.0 * (1 - m))];
            occupancy = [:space => [0.0, 1.0, 4.0], :shrub_rain => [0.0,
 ↪  0.0, 3.0 * m]],
            recruitment = [:seedling => ctx -> ctx.pool.shrub_rain / 2])
end
```

```
shrub (generic function with 1 method)
```

Two levels of wiring

Within a patch the two species share `space`, which stays inside the patch, and each exposes its seed rain:

```
patch_diagram = @relation (grass_rain, shrub_rain) begin
    grass(space, grass_rain)
    shrub(space, shrub_rain)
end
```

Box	name
1	grass
2	shrub

Port	box	junction
1	1	3
2	1	1
3	2	3
4	2	2

OuterPort	outer_junction
1	1
2	2

Junction	variable
1	grass_rain
2	shrub_rain
3	space

Between patches, the seed rains are joined:

```
meta_diagram = @relation (grass_rain, shrub_rain) begin
    A(grass_rain, shrub_rain)
    B(grass_rain, shrub_rain)
end
```

Box	name
1	A
2	B

Port	box	junction
1	1	1
2	1	2
3	2	1
4	2	2

OuterPort	outer_junction
1	1
2	2

Junction	variable
1	grass_rain
2	shrub_rain

```
patch(p; m) = compose(patch_diagram, [grass(p; m), shrub(p; m)])
metacommunity(m) = compose(meta_diagram, [patch(:A; m), patch(:B; m)])
meta = metacommunity(0.05)
species_names(meta), pool_names(meta)
```

```
([:grass_A, :shrub_A, :grass_B, :shrub_B], [:grass_rain, :shrub_rain])
```

The `space` pools do not appear among the metacommunity's pools: each belongs to one patch and nothing outside can read it.

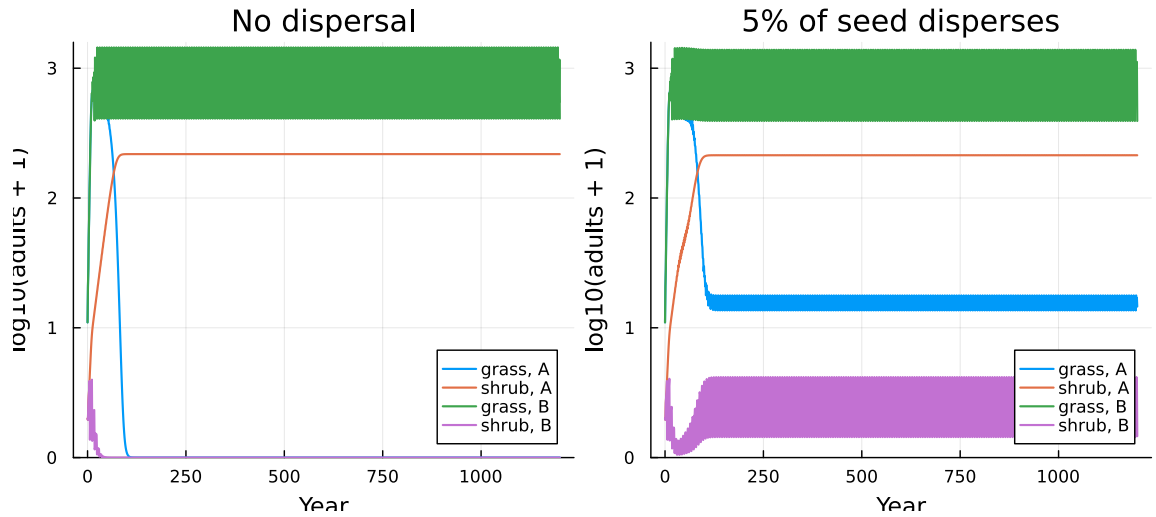
Isolated patches lose species; connected ones keep them

Patch `B` burns every sixth year.

```
envs = [(A = false, B = t % 6 == 0) for t in 1:1200]
n0 = [50.0, 10.0, 5.0, 0.0, 1.0, 50.0, 10.0, 5.0, 0.0, 1.0]

isolated = project(metacommunity(0.0), n0, envs)
connected = project(meta, n0, envs)

adults(sol) = [sol[:grass_A][:, 2] sol[:shrub_A][:, 3] sol[:grass_B][:, 2]
 ↪  sol[:shrub_B][:, 3]]
labels = ["grass, A" "shrub, A" "grass, B" "shrub, B"]
plot(plot(isolated.t, log10.(adults(isolated) .+ 1); label = labels, title =
 ↪  "No dispersal", lw = 1.5),
     plot(connected.t, log10.(adults(connected) .+ 1); label = labels, title
      ↪  = "5% of seed disperses", lw = 1.5);
     layout = (1, 2), xlabel = "Year", ylabel = "log10(adults + 1)", ylims
 ↪  = (0, 3.2), size = (800, 350),
     legend = :bottomright)
```



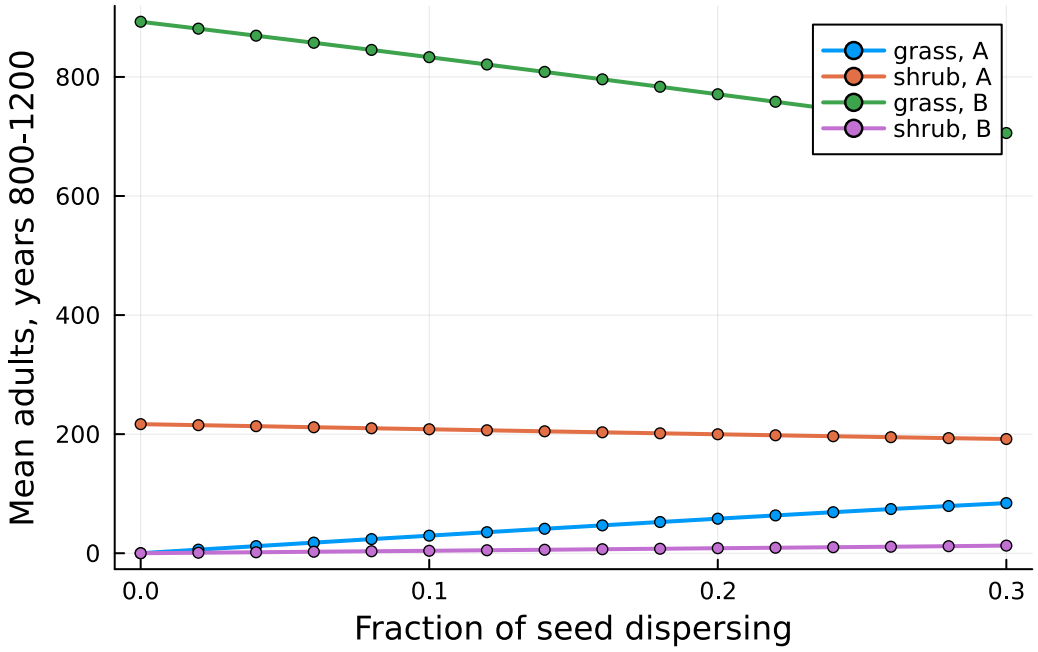

```
late(sol) = round.(vec(mean(adults(sol)[801:end, :]; dims = 1)); digits = 1)
(isolated = late(isolated), connected = late(connected))
```

```
(isolated = [0.0, 216.8, 892.7, 0.0], connected = [14.9, 212.5, 863.4, 2.1])
```

Without dispersal the shrub excludes the grass in the unburnt patch and fire excludes the shrub in the burnt one. With 5% of seed dispersing, each species is maintained in the patch where it would otherwise disappear, by seed arriving from the patch where it thrives.

Dispersal as a dial

```
ms = 0.0:0.02:0.3
means = reduce(vcat, [late(project(metacommunity(m), n0, envs))' for m in
 ↪  ms])
plot(ms, means; label = labels, xlabel = "Fraction of seed dispersing",
 ↪  ylabel = "Mean adults, years 800–1200",
     lw = 2, marker = :circle, ms = 3)
```



Reference

Lytle, D. A. & Tonkin, J. D. (2023). Matrix community models for ecology and evolution. *npj Biodiversity* 2, 26.